

Surname	Centre Number	Candidate Number
First name(s)		2

**GCE AS/A LEVEL**

2410U20-1



Z22-2410U20-1

FRIDAY, 27 MAY 2022 – AFTERNOON**CHEMISTRY – AS unit 2****Energy, Rate and Chemistry of Carbon Compounds**

1 hour 30 minutes

Section A**Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.

Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q9(c)**.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 5.	10	
6.	15	
7.	17	
8.	10	
9.	12	
10.	16	
Total	80	

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SECTION AAnswer **all** questions.

1. Bromine water can be used to test for alkenes.

(a) (i) State the expected colour change for a positive test for alkenes. [1]

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(ii) Draw the structure of the product formed when propene reacts with bromine water. [1]

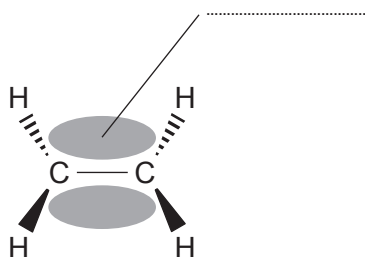
(b) Identify another reagent that can be used to test for the presence of alkenes. [1]

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2. Bonds in hydrocarbons are formed by the overlap of orbitals between each atom.

(a) Draw an *s*-orbital and a *p*-orbital in the space below. [1]

(b) Name the type of bond shown in the diagram below. [1]



3. A student suspects an unlabelled organic liquid is a carboxylic acid. Name the reagent(s) that must be added to the unknown organic liquid to test for the presence of a carboxylic acid. Give the expected observations for a positive result. [2]

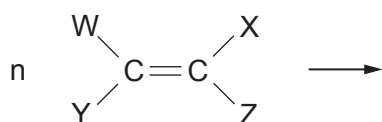
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4. Complete the equation below to show the product of addition polymerisation. [1]



5. State the meaning of the term 'standard enthalpy change of formation'. [2]

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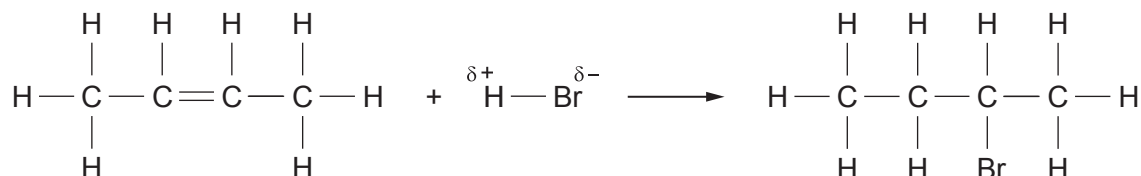


SECTION B

Answer **all** questions.

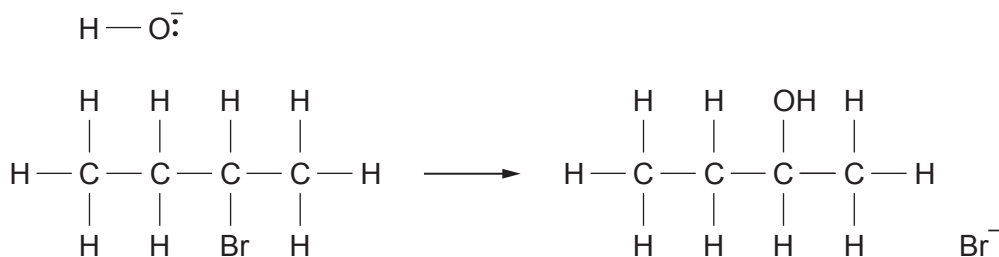
6. Butanone can be prepared from but-2-ene using a three-step synthesis.

(a) In the first step, but-2-ene is reacted with HBr to form 2-bromobutane.

(i) Circle the species that represents the electrophile. [1]

(ii) Name the type of bond fission that takes place in the H — Br bond in the first step of the mechanism. [1]

(b) In the second step, 2-bromobutane undergoes nucleophilic substitution to form butan-2-ol.



(i) Use curly arrows to complete the equation to show the mechanism of the nucleophilic substitution. Include any relevant partial charges. [2]

(ii) Give the reagents and conditions required for this nucleophilic substitution. [2]

(iii) State the classification of alcohol to which butan-2-ol belongs. [1]



- (c) In the final step, butan-2-ol is heated with acidified potassium manganate(VII) to produce butanone.

(i) State the role of the acidified potassium manganate(VII) in this reaction. [1]

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(ii) Explain why butanone can be removed from the reaction as it is formed using distillation, leaving unreacted butan-2-ol in the reaction mixture.

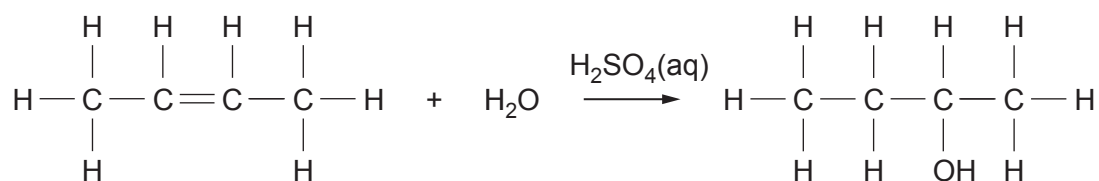
[2]

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- (d) Butan-2-ol can also be made directly by hydration of but-2-ene in the presence of dilute sulfuric acid, which acts as a catalyst.



(i) Suggest why the overall yield of the two-step synthesis is likely to be lower than the yield of the direct hydration. [1]

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- I. Name the type of reaction used to form alkenes from alcohols. [1]

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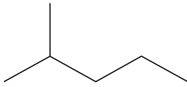
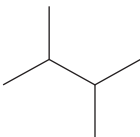
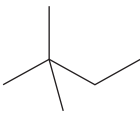
7. Petroleum ether (50–70) is a mixture of different alkanes extracted from crude oil which is commonly used as an organic solvent. The major components of petroleum ether (50–70) are the structural isomers of C_6H_{14} .

(a) (i) Give the meaning of the term 'structural isomer'. [1]

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(ii) Complete the table below showing important information about the isomers of C_6H_{14} . [3]

Name	Shortened structural formula	Skeletal formula	Boiling temperature / °C
hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$		69
2-methylpentane			62
3-methylpentane	$CH_3CH_2CH(CH_3)CH_2CH_3$		63
	$(CH_3)_2CHCH(CH_3)_2$		58
2,2-dimethylbutane	$CH_3C(CH_3)_2CH_2CH_3$		50



- (iii) State the relationship between the boiling temperature and the carbon chain length. Explain this relationship in terms of intermolecular forces. [2]

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- (b) Hexane can be used as a fuel in a combustion reaction.

- (i) Write an equation for the complete combustion of hexane in excess oxygen. [2]

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- (ii) The enthalpy change of combustion ($\Delta_c H^\theta$) for hexane is approximately $-4160 \text{ kJ mol}^{-1}$. Explain why the enthalpy change of combustion for the isomers of hexane should be similar. [2]

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- (iii) 2,2-dimethylbutane is the isomer of C_6H_{14} which ignites most readily. Suggest a reason for this. [1]

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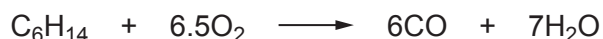
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- (iv) When hexane burns in a limited supply of oxygen it undergoes a different reaction known as incomplete combustion:



The bond enthalpy values for the bonds present in these molecules are given below:

Bond	Average bond enthalpy / kJ mol^{-1}
C — C	348
C — H	413
O = O	495
C $\equiv$ O (in CO)	1072
O — H	464

- I. Using a Hess cycle or otherwise, calculate the enthalpy change of this reaction.

[3]

enthalpy change = kJ mol^{-1}

- II. **Use the enthalpy values** from parts (b)(ii) and (b)(iv) I. to explain quantitatively why it is important to maintain an excess of oxygen while burning hexane as a fuel.

[2]

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III. State a health hazard associated with the incomplete combustion of hexane. [1]

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8. Compound **A** contains only carbon, hydrogen and an unknown halogen.

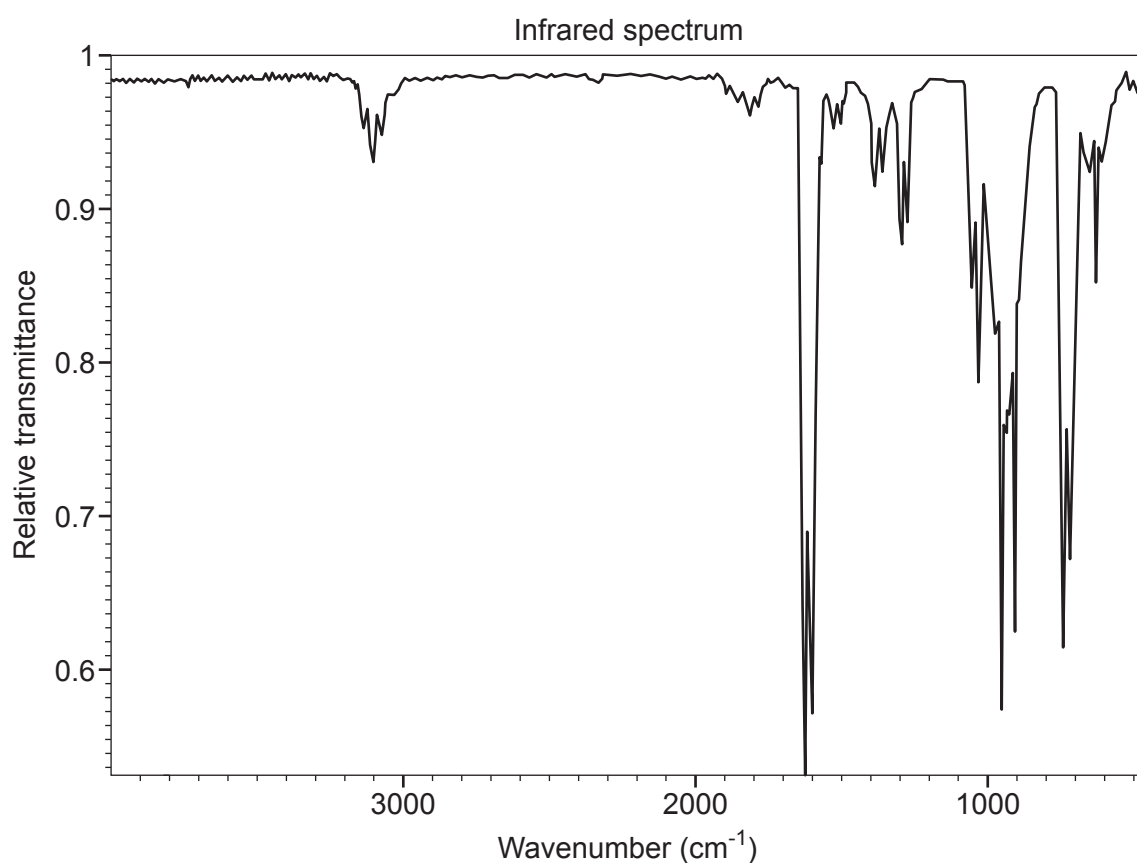
Refluxing compound **A** in aqueous sodium hydroxide followed by the addition of nitric acid and aqueous silver nitrate produces a white precipitate.

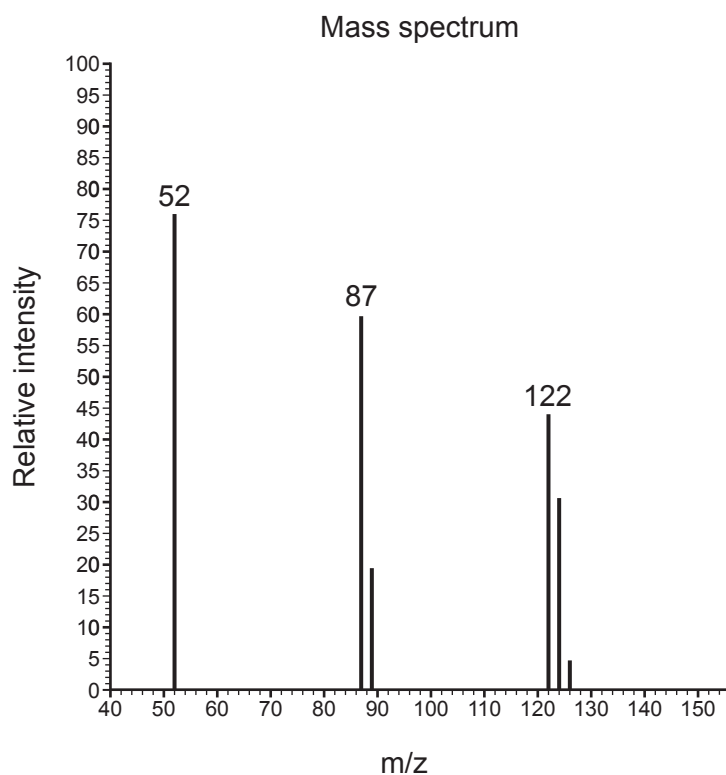
Elemental analysis of compound **A** indicates it contains 39.02% carbon and 3.25% hydrogen by mass.

When bromine is added to compound **A**, 123 g of compound **A** reacts with 320 g of bromine.

The ^1H NMR spectrum of compound **A** consists of only one peak. The ^{13}C NMR spectrum of compound **A** consists of two peaks.

The infrared spectrum and simplified mass spectrum are shown below and overpage.





Use **all** of the data provided to determine the identity of compound **A**. Explain your answer using information from all of the data sources provided.

[10]

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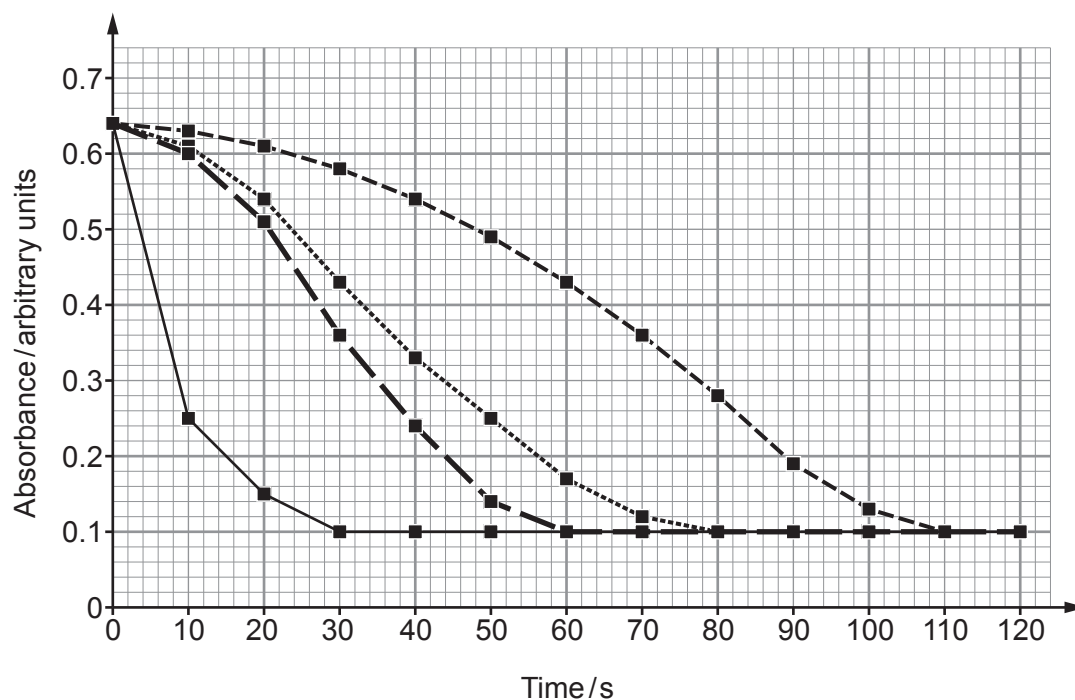
9. Chloe was investigating the effect of using catalysts on the rate of reaction.

She added 50 cm^3 of 0.1 mol dm^{-3} iron(III) nitrate solution to 50 cm^3 of 0.2 mol dm^{-3} sodium thiosulfate solution. The reaction forms a deep violet iron(III) complex which is unstable and is gradually reduced to form a light green iron(II) complex.

Chloe monitored the rate of reaction by measuring the absorption of light at a wavelength of 500 nm every 10 seconds for two minutes using a data logger.

- (a) The violet complex appears black at the beginning of the reaction. State the name of the technique used to monitor the rate of reaction by measuring the absorption of light. [1]

- (b) Chloe repeated the experiment three times adding 1 cm^3 of a different catalyst each time at a concentration 0.10 mol dm^{-3} . Below is a graph showing her results:



Key: - - - - - $\blacksquare$ - - - - - No catalyst - - - - - $\blacksquare$ - - - - - Fe^{2+}
 $\blacksquare$ Co^{2+} - - - - - $\blacksquare$ - - - - - Cu^{2+}

- (i) State which catalyst is the most effective. Explain your answer. [2]

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- (ii) Calculate the initial rate of reaction for the reaction catalysed by the copper(II) ions. [2]

rate = s^{-1}

- (iii) Each catalysed reaction contained the same number of moles of catalyst at the beginning of the reaction. Calculate the moles of catalyst left at the end of the reaction. [1]

moles = mol



- (c) Increasing temperature and the addition of a catalyst are two ways of increasing the rate of a reaction.

Using your knowledge of the Boltzmann distribution and particle theory, explain how the rate of reaction is increased using these two different methods.

You may use a diagram(s) to support your answer.

[6 QER]

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10. The crystallisation of sodium ethanoate from a super-saturated solution is used to release heat in reusable hand warmers.

- (a) A super-saturated solution of sodium ethanoate was made by dissolving 320 g of hydrated sodium ethanoate ($\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$) in 60 cm^3 of hot water. It was then allowed to cool to room temperature, which was measured as 17°C .

A thermometer was added to the solution, which caused the sodium ethanoate to start crystallising. The temperature of the process was recorded every 30 seconds for 3 minutes. The results are shown below:

Time/s	Temperature/ $^\circ\text{C}$
0	17
30	27
60	35
90	41
120	40
150	39
180	38

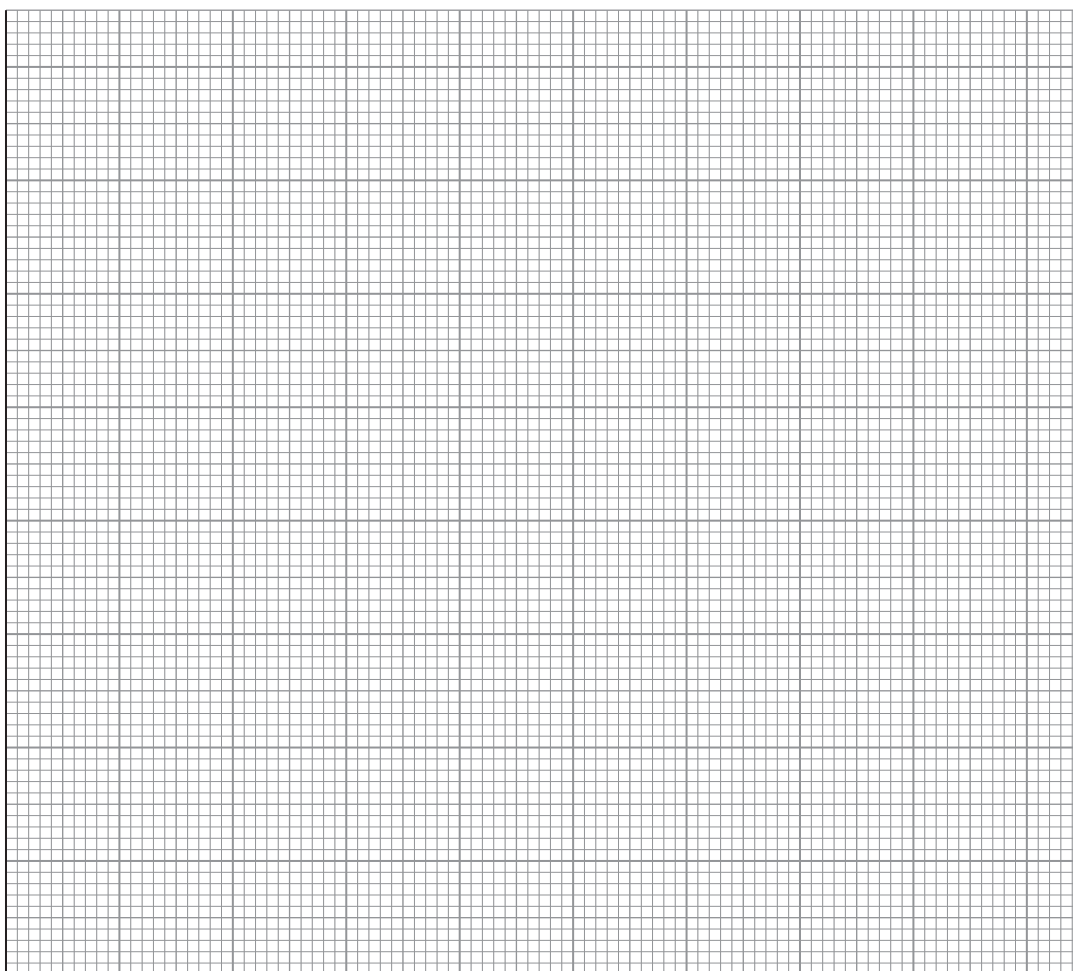


- (i) Plot the results on the graph paper below.

[2]

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Temperature / °C



Time / s

- (ii) Use your graph to calculate the maximum temperature change for this crystallisation.

[2]

maximum temperature change = °C



- (iii) Use the **total mass** of the sodium ethanoate solution and the temperature change from the graph to calculate the enthalpy change of crystallisation per mole of sodium ethanoate. Assume the density of water is 1.00 g cm^{-3} and the specific heat capacity of sodium ethanoate solution is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$.

$$M_r(\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}) = 136 \quad [4]$$

enthalpy change = kJ mol^{-1}

- (iv) Suggest a reason why the experimental enthalpy change is often lower than the theoretical enthalpy change. [1]

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- (b) Sodium ethanoate can be made in a neutralisation reaction. Complete the following equation: [2]



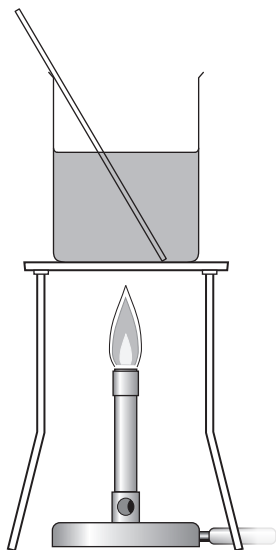
- (c) The carboxylic acid used to produce sodium ethanoate can be produced using an oxidation reaction.

- (i) Name the reagents and give the expected observations. [2]

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- (ii) A student proposed that the apparatus below should be used to perform this oxidation reduction experiment.



The teacher said that this would not work and would be unsafe. Draw a labelled diagram of the apparatus that should be used in this experiment. [3]

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